

Bangladesh University of Engineering and Technology
Department of Computer Science and Engineering

CSE 402: Numerical Analysis, Simulation and Modelling

Course Project — Final Report

Robust Newton–Raphson Power Flow Analysis for Singular and Ill-Conditioned Systems

A comparative study of SVD (Moore–Penrose), rank-revealing QR and Tikhonov regularization for singular Jacobians in unbalanced three-phase distribution feeders

Section B

Group Members (Student IDs)

2105094
2105102
2105104
2105116
2105117

Base paper: J. Jang, D. Kim and I.-Y. Kim, “Singularity Handling for Unbalanced Three-Phase Transformers in Newton–Raphson Power Flow Analyses Using Moore–Penrose Pseudo-Inverse,” *IEEE Access*, 2023.

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Abstract

Ungrounded (floating) three-phase transformer connections such as Yg–Delta and Delta–Delta can make the Jacobian of a Newton–Raphson (NR) power-flow solver singular or severely ill-conditioned, so that the usual direct linear solve fails. The base paper handles this with a Moore–Penrose pseudo-inverse computed by SVD. In this project we implemented a shared three-phase NR solver and compared four ways of solving the linear system in each iteration: a guarded direct solve, the SVD pseudo-inverse, rank-revealing QR (RRQR) and Tikhonov regularization. We tested them on reconstructed IEEE 4-, 13-, 37- and 69-bus feeders and on an IEEE 118-bus runtime benchmark. We checked the analytical Jacobian against finite differences and compared the feeder voltages with OpenDSS. We found that no single method works best everywhere. On the 13-bus feeder only Tikhonov converges at nominal load, because the SVD and RRQR steps oscillate. On the 37-bus feeder the opposite happens: SVD and RRQR converge and Tikhonov does not. A symmetrical-component study also shows that load imbalance on the floating 13-bus network mainly changes the zero-sequence voltage.

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1 Introduction

1.1 Background and Motivation

Power-flow analysis computes the voltage at every bus of a power network from given loads and generation. The resulting equations are nonlinear and are almost always solved by the Newton–Raphson (NR) method, which solves a linear system with the Jacobian matrix J at every iteration.

In unbalanced three-phase distribution feeders, some transformer connections (Delta windings and ungrounded wye) block zero-sequence current. If such a transformer leaves part of the network with no path to ground, the common-mode voltage in that part is not fixed by the network equations. The Jacobian then becomes exactly singular, or so close to singular that floating-point arithmetic cannot tell the difference, and the standard direct solve fails.

1.2 Base Paper

Jang, Kim and Kim [1] studied this problem. They replaced the direct solve with a Moore–Penrose pseudo-inverse computed by singular value decomposition (SVD) and truncated the near-zero singular values. They tested the idea on IEEE 4-, 13-, 37- and 69-bus feeders. They reported that the 13-bus case had a much larger voltage error than the other feeders (about 2×10^{-4} pu) but did not fully explain why, and they did not compare the pseudo-inverse with other numerical remedies.

1.3 Objectives

The project had three objectives:

1. Reproduce the base paper’s failure mode, in which the direct solver breaks down on singular or ill-conditioned Jacobians.
2. Compare the SVD pseudo-inverse with two alternatives, rank-revealing QR and Tikhonov regularization, inside the same NR solver.
3. Look for a structural explanation of the large error on the 13-bus feeder.

1.4 Scope

All four solvers use exactly the same NR loop, network models and convergence tolerance, so any difference between them comes from the linear solve alone. The feeders were rebuilt from official public data and checked in two ways: the Jacobian against finite differences and the voltages against OpenDSS [2]. We use the IEEE 118-bus system only to measure how runtime grows with system size.

2 Theoretical Background

2.1 Newton–Raphson Power Flow

The state vector x contains the rectangular voltage components $e = \text{Re}(V)$ and $f = \text{Im}(V)$ of every non-slack bus-phase. The mismatch function $F(x)$ is the difference between the calculated

and the specified real and reactive power injection at each bus-phase. NR iterates [3]

$$J_k \Delta x_k = -F(x_k), \quad x_{k+1} = x_k + \Delta x_k, \quad J_k = \left. \frac{\partial F}{\partial x} \right|_{x_k}, \quad (1)$$

and stops when $\|F(x_k)\|_\infty$ is below a tolerance. The linear solve in (1) is the only numerically delicate step and the only step that differs between the methods compared here.

2.2 Singularity and Ill-Conditioning

The condition number $\kappa_2(J) = \sigma_{\max}/\sigma_{\min}$ measures how much errors in the data can be amplified in the solution. An LU-based solver reports an error only when a pivot is exactly zero, which almost never happens in floating point. For a near-singular matrix it returns an answer whose round-off error has been amplified by about $\kappa_2(J)$. The residual $\|J\Delta x + F\|$ can still look small, because it only checks that Δx satisfies the linear system, not that it is close to the true solution [4].

2.3 Symmetrical Components

The Fortescue transform [5] splits a set of phase voltages into zero-, positive- and negative-sequence components:

$$\begin{bmatrix} V_0 \\ V_1 \\ V_2 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 1 & 1 & 1 \\ 1 & a & a^2 \\ 1 & a^2 & a \end{bmatrix} \begin{bmatrix} V_a \\ V_b \\ V_c \end{bmatrix}, \quad a = e^{j2\pi/3}. \quad (2)$$

In a floating region, V_0 is exactly the component that the network equations do not determine, so it is a natural diagnostic for this problem.

3 Methodology

3.1 Solver Architecture

Every experiment uses one shared NR loop. The linear-solve step is chosen by a `method` argument, and all four solvers return the same `(dx, diagnostics)` pair. The analytical Jacobian was checked against centered finite differences with the relative error

$$E_J = \frac{\|J^{\text{analytic}} - J^{\text{FD}}\|_F}{\|J^{\text{FD}}\|_F}, \quad J_{ij}^{\text{FD}} = \frac{F_i(x + he_j) - F_i(x - he_j)}{2h}. \quad (3)$$

The check uses a perturbed operating point rather than the flat start, so that errors that appear only when the mismatch is non-zero are not hidden. No result was used until $E_J < 10^{-9}$ held for that feeder.

3.2 Linear Solvers Compared

Table 1 summarizes the four strategies.

Table 1: The four linear-solve strategies used inside the shared NR loop.

Method	Computation	Notes
Direct	LU via <code>numpy.linalg.solve</code>	Rejects the step if $\kappa_2(J) > 10^{12}$, because LU on a near-singular J gives an unreliable answer without reporting an error.
SVD (base paper) [6]	$\Delta x = V\Sigma^+U^H$ rhs, with $\Sigma_{ii}^+ = 1/\sigma_i$ when $\sigma_i > \tau$ and 0 otherwise	$\tau = \max(\text{atol}, \text{rtol} \cdot \sigma_{\max})$; every truncated singular value is logged.
RRQR [7]	Pivoted QR $JP = QR$ for rank diagnostics, then LAPACK GELS	Complete orthogonal factorization giving the minimum-norm least-squares solution; avoids naive back-substitution on a rank-deficient R .
Tikhonov [8]	$\min \ J\Delta x - \text{rhs}\ ^2 + \lambda\ \Delta x\ ^2$, solved as $\begin{bmatrix} J \\ \sqrt{\lambda}I \end{bmatrix} \Delta x \approx \begin{bmatrix} \text{rhs} \\ 0 \end{bmatrix}$	$\lambda = \alpha \sigma_{\max}^2$; the reported feeder comparisons explicitly use $\alpha = 10^{-8}$. The library’s conservative 10^{-12} default is not treated as a recommended value. $J^T J$ is never formed, since that would square κ_2 .

SVD and RRQR both compute the *exact* minimum-norm solution of the linearized step. Tikhonov instead *damps* the step, trading a small bias for stability. As Section 5 shows, this difference decides which method converges.

3.3 Test Feeders

The feeders were rebuilt from official sources [9, 10, 11] and modified as described in the base paper, for example by adding transformers or changing their connections. The 4-bus case was built in-house from the transformer parameters given in the base paper. Every modification and assumption is recorded in the project’s provenance file (`data/provenance.yml`). Table 2 lists the resulting systems.

Table 2: Reconstructed feeders: size, flat-start conditioning and Jacobian validation.

Feeder	n	$\kappa_2(J_0)$	Near-zero σ_i	E_J
IEEE 4-bus (Case B)	18	61.5 [†]	0	$< 10^{-9}$
IEEE 13-bus	64	machine-singular	2	8.6×10^{-11}
IEEE 37-bus (radial)	222	machine-singular	4	7.8×10^{-11}
IEEE 37-bus (quasi-radial)	222	machine-singular	4	4.6×10^{-11}
IEEE 69-bus	408	2.28×10^5	0	7.0×10^{-11}

n : number of real unknowns. Near-zero: $\sigma_i < 10^{-6}\sigma_{\max}$ at flat start. “Machine-singular” means $\kappa_2 \sim 10^{17}$ – 10^{18} , set by round-off. [†]Grows past 10^{12} during the iterations.

3.4 Validation against OpenDSS

Each converged solution was compared with an OpenDSS solution of the same circuit, built from the same source data. For the 4- and 69-bus feeders both programs solve the same mathematical model, so the comparison is a true validation. For the 13- and 37-bus feeders our solver approximates delta loads by equivalent wye loads, so those comparisons also include model differences.

This check found two real per-unit bugs. On the 13-bus feeder the loads were $3\times$ too light: they had been normalized by the three-phase base instead of the per-phase base. On the 69-bus feeder

the loads were $3\times$ too heavy: each bus’s total three-phase load had been applied to every phase. We found both bugs by first comparing a small, well-conditioned sub-circuit with OpenDSS. On the full ill-conditioned system, a scaling bug looks very much like the normal disagreement caused by the null space. For example, the 69-bus bug showed up only as a plausible-looking 33% error at one bus.

4 Implementation

The project is written in Python with NumPy and SciPy. OpenDSS was driven through `openssdirect` and `pandapower` [12] provided the 118-bus case. Table 3 shows the repository layout.

Table 3: Repository organization.

Directory	Contents
<code>src/model/</code>	Bus-phase state, Y_{bus} assembly, transformer stamping, feeder data
<code>src/powerflow/</code>	Mismatch $F(x)$, analytical Jacobian, shared NR loop
<code>src/solvers/</code>	<code>direct.py</code> , <code>svd_pinv.py</code> , <code>rrqr.py</code> , <code>tikhonov.py</code>
<code>src/diagnostics/</code>	Symmetrical-component analysis
<code>src/validation/</code>	OpenDSS reference pipeline and cross-checks
<code>experiments/</code>	One standalone script per feeder or study
<code>tests/</code>	<code>pytest</code> suite (86 tests)
<code>data/</code>	Raw feeder files and <code>provenance.yml</code>

The test suite covers the finite-difference Jacobian checks, each solver on controlled rank-deficient matrices, dispatch through the NR loop, feeder-level structural properties (such as the number of near-zero singular values) and the sequence-component code. All 86 tests pass on Python 3.13. On Windows the OpenDSS backend can print native error messages under `pytest`, so we ran the OpenDSS comparisons as standalone scripts.

5 Results

5.1 Summary of Solver Outcomes

Table 4 shows whether each solver converged on each feeder at nominal load. Sections 5.2–5.9 discuss each feeder.

Table 4: Convergence at nominal load ($\|F\|_{\infty} < 10^{-9}$). The number of iterations is given in parentheses; $\times$ means the solver failed.

Feeder	Direct	SVD	RRQR	Tikhonov ($\alpha = 10^{-8}$)
IEEE 4-bus Case A (loaded)	✓ (8)	✓	✓	✓
IEEE 4-bus Case B (no load)	$\times$ guard	✓	✓	✓
IEEE 13-bus	$\times$ guard	$\times$ oscillates	$\times$ oscillates	✓ (15)
IEEE 37-bus (radial)	$\times$ guard	✓ (10)	✓ (10)	$\times$
IEEE 37-bus (quasi-radial)	$\times$ guard	✓ (10)	✓ (10)	$\times$
IEEE 69-bus	✓ (7)	✓ (7)	✓ (7)	$\times$

“Guard”: rejected because $\kappa_2(J) > 10^{12}$. “Oscillates”: $\|F\|_{\infty}$ does not settle within the iteration limit (60–80).

Tikhonov failures reached the 80-iteration limit.

5.2 IEEE 4-Bus: Reproducing the Failure

With the base paper’s Yg–Delta transformer (6 MVA, 12.47/4.16 kV, $R = 0.01$, $X = 0.06$ pu) and no load on the Delta secondary (Case B), κ_2 rises from 61.5 at flat start to 1.80×10^{12} at the seventh iteration (Figure 1), and the direct solver’s guard stops the run. This reproduces the qualitative failure in the base paper. Their reported $\kappa_2 \sim 10^{160}$ comes from a different topology that we did not try to match exactly. SVD, RRQR and Tikhonov all converge on Case B. With a loaded secondary (Case A), κ_2 stays below 2.6×10^4 , and the solution agrees with OpenDSS to 1.49×10^{-6} pu.

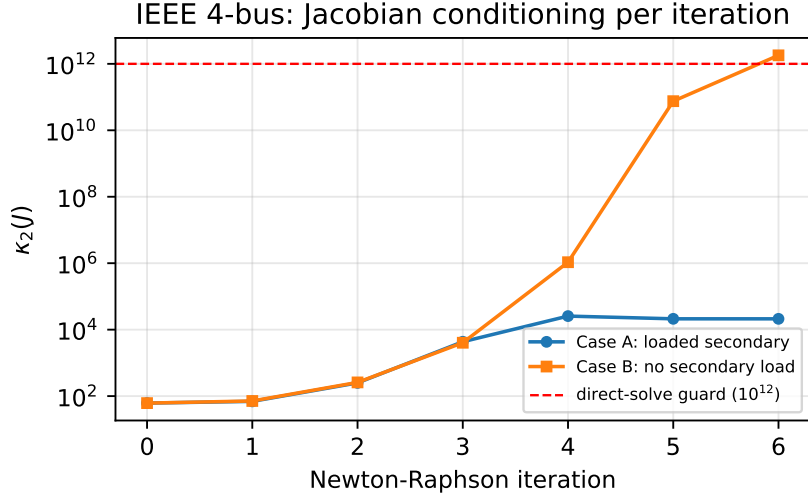


Figure 1: IEEE 4-bus: Jacobian condition number at each NR iteration. Case B crosses the direct-solve guard; Case A stays far below it.

5.3 IEEE 13-Bus: Oscillation of Minimum-Norm Steps

The base paper changes the substation transformer T1 to Yg–Delta and the service transformer T2 to Delta–Delta. This leaves the whole downstream network without a ground reference. The flat-start Jacobian is machine-singular with exactly two near-zero singular values, as the structure predicts. One common-mode direction comes from T1’s floating secondary, which spans every bus except the source. The second comes from bus 634 behind T2, because the zero-sequence transfer block of a Delta–Delta transformer is exactly zero.

The direct solver fails by construction. More surprisingly, SVD and RRQR also fail at and above nominal load, although both converge at $0.5\times$ and $0.75\times$ nominal (Figure 2). Their undamped minimum-norm steps oscillate instead of reducing $\|F\|_\infty$. Only Tikhonov converges at nominal load (15 iterations), and it keeps converging up to $1.25\times$ nominal. On this feeder, damping the step is therefore necessary for convergence, not just a way to handle the singularity.

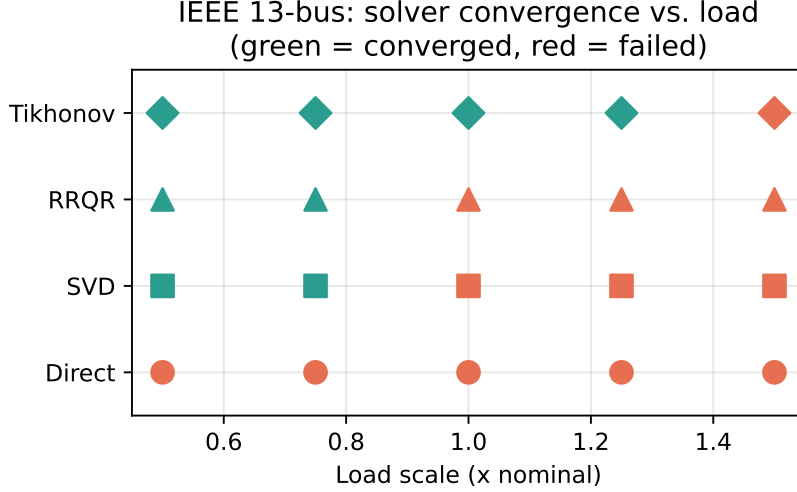


Figure 2: IEEE 13-bus: solver convergence against load scale. SVD and RRQR fail from 1.0 $\times$ nominal upward. Tikhonov converges up to 1.25 $\times$ and fails at 1.5 $\times$.

5.4 IEEE 13-Bus: Tikhonov Parameter Sensitivity

The nominal-load result is sensitive to the regularization strength; it is not evidence that $\alpha = 10^{-8}$ is a universal default. We held the feeder, flat start, tolerance and 80-iteration budget fixed and varied only α over the project grid. Table 5 shows that 10^{-8} is the only tested value that converges. Smaller values leave too little damping to stop the oscillation, while larger values over-damp the Newton corrections and stall. The dependence is therefore non-monotonic.

Table 5: IEEE 13-bus nominal-load sensitivity to the Tikhonov parameter.

α	Converged	Iterations	Final $\ F\ _\infty$
0	×	80	1.93×10^{-2}
10^{-12}	×	80	1.64×10^{-2}
10^{-10}	×	80	1.50×10^{-2}
10^{-8}	✓	15	2.02×10^{-10}
10^{-6}	×	80	1.80×10^{-3}
10^{-4}	×	80	1.22×10^{-2}
10^{-2}	×	80	3.22×10^{-1}

5.5 IEEE 13-Bus: Zero-Sequence Analysis

Using (2), we added extra unbalance on top of the feeder’s existing unbalanced loads, scaling phase a by $1 + L$ and phase c by $1 - L$ for L from 0 to 30%. We then tracked the mean $|V_0|$ and $|V_1|$ over the three-phase buses. Over this sweep $|V_0|$ rises from 0.337 to 0.408 pu (+21%), while $|V_1|$ changes by less than 1% (0.807 to 0.800 pu, Figure 3). The extra unbalance therefore goes almost entirely into the component that the equations leave undetermined, not into the well-determined positive-sequence component.

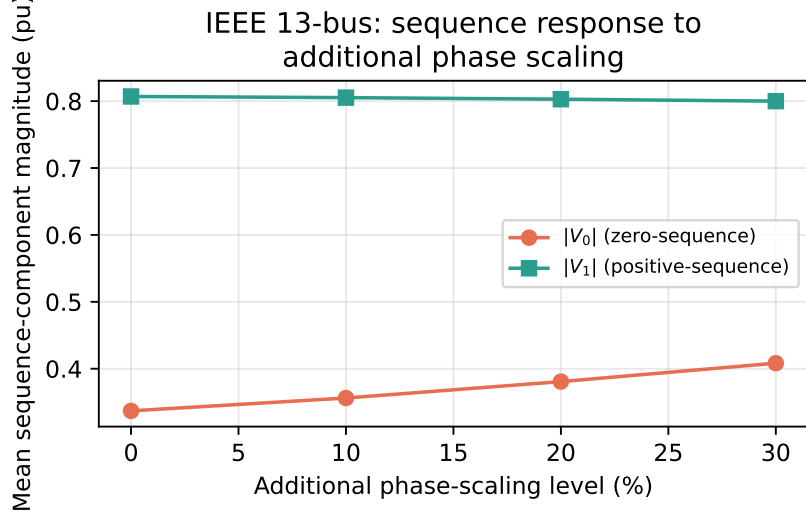


Figure 3: IEEE 13-bus: mean $|V_0|$ and $|V_1|$ over a 0–30% extra phase-imbalance sweep.

Replacing the Delta windings with grounded wye confirms that the floating windings cause this behavior (Figure 4). With both transformers grounded, $\kappa_2(J_0)$ falls from machine-singular to about 773 and the mean $|V_0|$ falls from 0.337 to 0.066 pu. Grounding one transformer at a time separates their effects. Grounding T2 alone removes the extreme conditioning, while grounding T1 alone gives most of the reduction in $|V_0|$.

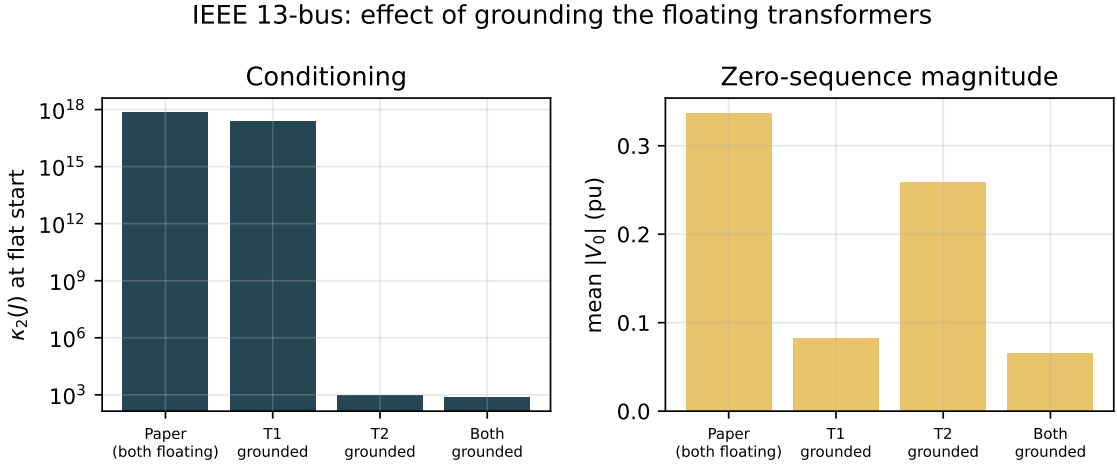


Figure 4: IEEE 13-bus: flat-start conditioning (left) and mean $|V_0|$ (right) when T1 and/or T2 are grounded.

5.6 IEEE 37-Bus: The Reversal

The official 37-bus feeder already has Delta–Delta substation and service transformers, and the service transformer carries no load. The flat-start Jacobian is again machine-singular, this time with four near-zero singular values. We confirmed this count numerically but could not fully derive it; the unloaded leaf bus behind the service transformer seems to contribute more than one direction.

Despite this, SVD and RRQR converge in 10 iterations without oscillation. Tikhonov, with the same $\alpha = 10^{-8}$ that worked on the 13-bus feeder, fails to converge within 80 iterations. **This is the reverse of the 13-bus result and the main finding of the project: no single**

solver, and no single fixed regularization strength, is best on every feeder. The size of the numerical null space and the value of κ_2 do not predict which method will converge. Adding the three quasi-radial loop lines changes the large singular values, but the near-zero count and the convergence pattern stay the same (Table 4).

5.7 IEEE 69-Bus: Ill-Conditioned but Not Singular

We modified the 69-bus feeder by replacing branches 3–4, 3–28 and 3–36 with Yg–Delta transformers. Here even the direct solver converges: $\kappa_2 = 2.28 \times 10^5$ at flat start and up to 5.7×10^{10} during the iterations, still below the guard. Y_{bus} still has an exact null direction behind each transformer. A uniform voltage shift over the 47-bus region behind one transformer leaves every branch current unchanged to within 10^{-13} . The NR Jacobian does not inherit this singularity, because its component along that direction is proportional to the current already flowing there, and this region carries most of the feeder’s load. Whether a floating topology makes the Jacobian exactly singular therefore depends on where the load sits, not only on the topology. Tikhonov at $\alpha = 10^{-8}$ again fails to converge on this feeder.

5.8 Cross-Validation with OpenDSS

Figure 5 and Table 6 summarize the comparison. The 4- and 69-bus feeders agree almost to machine precision. This confirms the network assembly and the per-unit conversion. The 13- and 37-bus feeders differ by a few percent at the median, with larger errors at buses aligned with the near-null space. Different regularizations, namely our Tikhonov solver and OpenDSS’s internal `PPM_Antifloat`, can choose different solutions within an undetermined subspace. However, the wye approximation of delta loads, the ideal voltage regulators and the omitted line charging also add to the difference, so we cannot attribute the difference to regularization alone.

Table 6: Voltage agreement with OpenDSS.

Feeder	Median error	Maximum error
IEEE 4-bus (Case A)*	—	1.5×10^{-6} pu
IEEE 13-bus	6.6%	26.5%
IEEE 37-bus (radial)	2.3%	6.5%
IEEE 37-bus (quasi-radial)	2.3%	6.1%
IEEE 69-bus	3.3×10^{-8}	8.2×10^{-6}

* Absolute phase-voltage magnitude error. The other rows give relative line-to-line voltage error.

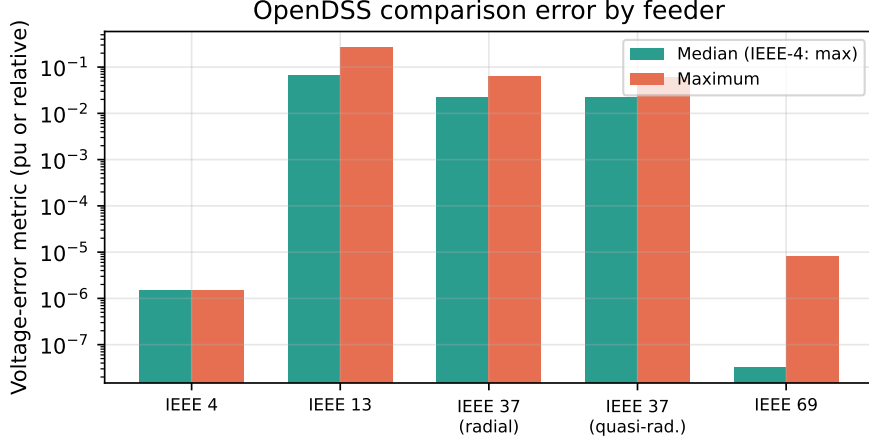


Figure 5: OpenDSS comparison error by feeder (log scale).

5.9 IEEE 118-Bus: Runtime Scaling

As in the base paper, we used a larger system only to measure computational cost, not to study singularity. We timed the four solvers on the admittance matrix of pandapower’s `case118`, repeated block-diagonally up to $8\times$, which gives real-valued systems from 236 to 1888 unknowns (Figure 6). The fitted runtime exponents are about 1.8–2.4 over this range. This is below the cubic asymptotic cost of dense factorizations because of fixed overheads and the threaded BLAS. Tikhonov is always the slowest, because its augmented system has twice as many rows. Direct is the fastest at larger sizes. Exact timings depend on the machine and the BLAS library.

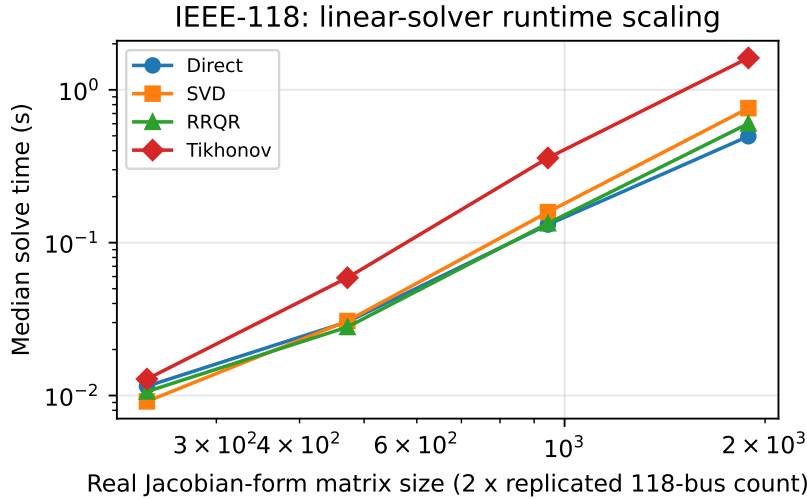


Figure 6: Solver runtime against matrix size on the replicated IEEE 118-bus system (log-log).

6 Discussion

The results show that the solver should be chosen to match how the Jacobian is rank-deficient, instead of using one method everywhere. The base paper’s pseudo-inverse works well when the Jacobian stays full rank despite a floating topology (69-bus, where it is not even needed). It also works when the iteration can tolerate an undamped minimum-norm step (37-bus). On the

13-bus feeder, however, the same step oscillates, and only the damped Tikhonov step converges. SVD and RRQR are not wrong there, since each one solves its linearized problem exactly. The sequence of those exact steps simply does not make progress on the nonlinear problem. This is a different failure from the singularity that the base paper addresses. RRQR shows the same oscillation as SVD, so switching from SVD to RRQR does not fix it. Tikhonov, in turn, failed on three feeders with a fixed α , which shows that the regularization strength has to be tuned to the problem.

The zero-sequence study suggests a partial answer to the base paper’s open question about the 13-bus feeder. On a floating, unbalanced network, a converged solution with a small residual is not necessarily *unique*. Different valid regularizations disagree exactly in the undetermined V_0 direction. The base paper measured its error against a single reference solver, so part of the reported error may come from this ambiguity rather than from the pseudo-inverse itself. Our reconstruction is independent of the base paper’s, so we do not claim to explain their 2.08×10^{-4} pu figure exactly. The mechanism we found, imbalance collecting in V_0 , is nevertheless a concrete and testable explanation.

7 Limitations

- **Model simplifications.** Delta loads are approximated by equivalent wye loads, voltage regulators are modeled as ideal fixed connections, and ungrounded-wye windings are not implemented. The code raises an error for them instead of using a wrong model.
- **No globalization.** The NR loop has no line search, trust region or adaptive damping, and Tikhonov uses a fixed α taken from a coarse sweep. Our conclusions apply to these implementations only.
- **Ambiguities in the base paper.**
 - The 13-bus case: the base paper says the modified case has seven loads, but the official data has eight load locations, and we kept all eight.
 - The 37-bus transformer reactance: the base paper’s value is 0.00181 pu, but the official value is 0.0181 pu, and we used the official value.
 - The 69-bus voltage base: the base paper gives 13.8 kV, but the network operates at 12.66 kV, and we used 12.66 kV.
 - The 69-bus transformers: we assumed they replace existing branches rather than being added in parallel.
- **Imbalance rule.** The phase-imbalance sweep uses a rule we defined ourselves, because the base paper does not give one. The 69-bus feeder was tested only in balanced form.

8 Conclusion

We built a shared three-phase Newton–Raphson solver and compared four linear-solve strategies on five reconstructed feeder variants. The main result is that no single method is best everywhere. The undamped minimum-norm step used by the base paper’s pseudo-inverse oscillates on the 13-bus feeder at realistic load. The Tikhonov regularization that works there fails on the 37-bus feeder, where SVD and RRQR converge easily. A symmetrical-component study showed that load imbalance on the floating 13-bus network goes mainly into the undetermined zero-sequence voltage, which partly explains that feeder’s large error. Finally, comparing small, well-conditioned sub-circuits with OpenDSS before trusting results on the full ill-conditioned systems found two real per-unit bugs.

Future work. Possible next steps include adding line search or trust-region globalization to the NR loop, choosing the Tikhonov α automatically (for example with the L-curve or discrepancy principle), modeling ungrounded-wye windings and delta loads exactly, and using sparse factorizations for large systems.

References

- [1] J. Jang, D. Kim, and I.-Y. Kim, “Singularity handling for unbalanced three-phase transformers in newton-raphson power flow analyses using moore-penrose pseudo-inverse,” *IEEE Access*, vol. 11, pp. 43041–43052, 2023.
- [2] R. C. Dugan and EPRI, “Reference Guide: The Open Distribution System Simulator (OpenDSS).” <https://sourceforge.net/projects/electricdss/>. Electric Power Research Institute, Accessed 2026.
- [3] W. F. Tinney and C. E. Hart, “Power flow solution by newton’s method,” *IEEE Transactions on Power Apparatus and Systems*, vol. PAS-86, no. 11, pp. 1449–1460, 1967.
- [4] G. H. Golub and C. F. Van Loan, *Matrix Computations*. Johns Hopkins University Press, 4th ed., 2013.
- [5] C. L. Fortescue, “Method of symmetrical co-ordinates applied to the solution of polyphase networks,” *Transactions of the American Institute of Electrical Engineers*, vol. XXXVII, no. 2, pp. 1027–1140, 1918.
- [6] R. Penrose, “A generalized inverse for matrices,” *Mathematical Proceedings of the Cambridge Philosophical Society*, vol. 51, no. 3, pp. 406–413, 1955.
- [7] T. F. Chan, “Rank revealing QR factorizations,” *Linear Algebra and its Applications*, vol. 88–89, pp. 67–82, 1987.
- [8] P. C. Hansen, *Rank-Deficient and Discrete Ill-Posed Problems: Numerical Aspects of Linear Inversion*. SIAM, 1998.
- [9] IEEE PES Distribution System Analysis Subcommittee, “Distribution Test Feeders.” <https://cmte.ieee.org/pes-testfeeders/resources/>. Accessed 2026.
- [10] R. D. Zimmerman, C. E. Murillo-Sánchez, and R. J. Thomas, “MATPOWER: Steady-state operations, planning, and analysis tools for power systems research and education,” *IEEE Transactions on Power Systems*, vol. 26, no. 1, pp. 12–19, 2011.
- [11] M. E. Baran and F. F. Wu, “Network reconfiguration in distribution systems for loss reduction and load balancing,” *IEEE Transactions on Power Delivery*, vol. 4, no. 2, pp. 1401–1407, 1989.
- [12] L. Thurner, A. Scheidler, F. Schäfer, J.-H. Menke, J. Dollichon, F. Meier, S. Meinecke, and M. Braun, “pandapower—an open-source python tool for convenient modeling, analysis, and optimization of electric power systems,” *IEEE Transactions on Power Systems*, vol. 33, no. 6, pp. 6510–6521, 2018.